

STAGE 0

Direct conditioning transfer (NIXL)

NIXL connector

STAGE 1

H3 request

prompt +
reference media

Text encoder

layer-50 residual stream

Payload producer

register conditioning buffer

Control plane

ZMQ key + memory metadata

Data plane

NIXL / UCX consumer READ

Diffusion model runner

receive conditioning

H3 DiT

denoising

Video VAE Audio VAE

MP4 output

hidden_states
 $[S \times 5120]$, BF16
token_tags $[S]$

reference media + request metadata

Legacy conditioning route (IPC + orchestration)

Engine Core IPC

serialize payload

Orchestrator

route payload

Stage 1 IPC

rebuild payload